

Section : Physics

Q.1 Two capacitors of capacitances $3\ \mu\text{F}$ and $6\ \mu\text{F}$ are charged to potentials of $2\ \text{V}$ and $3\ \text{V}$, respectively. Now, if they are connected in parallel, the charge on each of the two capacitors, will be

1. $6\ \mu\text{C}$ and $18\ \mu\text{C}$
2. $9\ \mu\text{C}$ and $12\ \mu\text{C}$
3. $8\ \mu\text{C}$ and $16\ \mu\text{C}$
4. $24\ \mu\text{C}$ and $24\ \mu\text{C}$

Options 1. 1

2. 2
3. 3
4. 4

Q.2 Arrange the following electromagnetic waves in decreasing order of their respective wavelengths.

- (A) UV-rays
- (B) X-rays
- (C) Radio waves
- (D) Visible light

Choose the correct answer from the options given below:

1. (A)>(B)>(C)>(D)
2. (B)>(A)>(C)>(D)
3. (D)>(A)>(C)>(B)
4. (C)>(D)>(A)>(B)

Options 1. 1

2. 2
3. 3
4. 4

Q.3 The magnetisation at the centre of a bar magnet (magnetic length $10\ \text{cm}$), having pole strength $6.3\ \text{A m}$, and area of cross-section $A = 0.70\ \text{cm}^2$ is

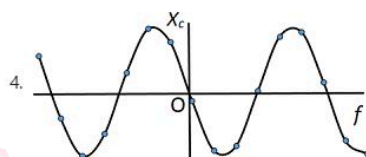
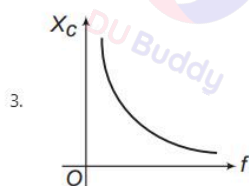
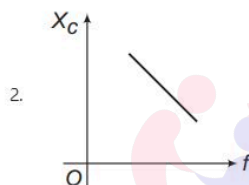
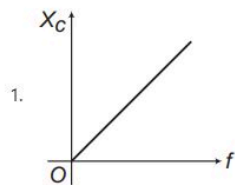
1. $1.1 \times 10^5\ \text{A m}^{-1}$
2. $9.0 \times 10^4\ \text{A m}^{-1}$
3. $9.0 \times 10^3\ \text{A m}^{-1}$
4. $9.0 \times 10^{-4}\ \text{A m}^{-1}$

Options 1. 1

2. 2
3. 3
4. 4

Q.4

Identify the graph which correctly represents the variation of capacitive reactance (X_C) with frequency (f)



Options 1. 1

2. 2

3. 3

4. 4

Q.5 Under which of the following conditions the power delivered by the source of circuit will be maximum

1. $\omega L = \omega C$

2. $\omega L = \omega C^2$

3. $\omega L = (\omega C)^{-1}$

4. $(\omega L)^{-1} = \omega C$

Options 1. 1

2. 2

3. 3

4. 4

Q.6 The impact parameter for an α -particle approaching a target nucleus is minimum when the scattering angle θ is

1. 0

2. $(\pi/4)$ rad

3. $(\pi/2)$ rad

4. (π) rad

Options 1. 1

2. 2

3. 3

4. 4

Q.7 A wheel with 10 metallic spokes each of length 1.0 m long is rotated with a speed of 240 rev/min in a plane normal to magnetic field of 0.4 G at the place, the induced emf between the axle and the rim of the wheel is:

Given that $1 \text{ G} = 10^{-4} \text{ T}$.

1. $6.28 \times 10^{-5} \text{ V}$

2. $7.536 \times 10^{-4} \text{ V}$

3. $7.536 \times 10^{-5} \text{ V}$

4. $5 \times 10^{-4} \text{ V}$

Options 1. 1

2. 2

3. 3

4. 4

Q.8 The physical quantity which is reciprocal of resistance is-

1. Conductivity

2. Resistivity

3. Conductance

4. Current density

Options 1. 1

2. 2

3. 3

4. 4

Q.9

In the circuit shown below, $C_1 = 6 \mu\text{F}$, $C_2 = 3 \mu\text{F}$ and battery $B = 40 \text{ V}$. Initially, switch A_1 is opened and the switch A_2 is closed. Now switch A_2 is opened and A_1 is closed. The charge on C_2 will be



1. $100 \mu\text{C}$

2. $40 \mu\text{C}$

3. $80 \mu\text{C}$

4. $18 \mu\text{C}$

Options 1. 1

2. 2

3. 3

4. 4

Q.10 Match List-I with List-II

List-I

Physical quantity

List-II

Unit

- (A) Magnetic moment (I) N A^{-2}
- (B) Magnetic field strength (II) N m rad^{-1}
- (C) Torsion constant (III) A m^{-1}
- (D) Magnetic permeability (IV) A m^2

Choose the correct answer from the options given below:

- (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
- (A) - (II), (B) - (I), (C) - (III), (D) - (IV)
- (A) - (III), (B) - (IV), (C) - (II), (D) - (I)
- (A) - (IV), (B) - (III), (C) - (II), (D) - (I)

Options 1. 1

2. 2
3. 3
4. 4

Q.11 An infinite line of charge has a linear charge density of 10^{-7} C m^{-1} . The magnitude of the force acting on a charge $6.4 \times 10^{-19} \text{ C}$ kept at a distance of 8 cm from the line of charge is

1. $7.2 \times 10^{-15} \text{ N}$
2. $7.2 \times 10^{-17} \text{ N}$
3. $7.2 \times 10^{-21} \text{ N}$
4. $1.44 \times 10^{-14} \text{ N}$

Options 1. 1

2. 2
3. 3
4. 4

Q.12 According to Joule's law of heating, the rate of heat produced in a resistor is

- directly proportional to the current for the given resistance
- directly proportional to the square of the current for the given resistance
- directly proportional to the resistance for a given value of current
- inversely proportional to the resistance for a given voltage

Choose the correct answer from the options given below:

- (A), (B) and (C) only
- (A), (C) and (D) only
- (A), (B), (C) and (D)
- (B), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.13 For a given convex lens, if the object distance is:

- $u = -\infty$
- $u = -2f$
- $u = -f$
- $u = -f/3$

The distance of the image is arranged in decreasing order:

Choose the correct answer from the options given below:

- (A), (B), (D), (C)
- (A), (B), (C), (D)
- (B), (A), (D), (C)
- (C), (B), (A), (D)

Options 1. 1

2. 2
3. 3
4. 4

Q.14 Match List-I with List-II

List-I

Physical quantity

(A) Width of a central maximum

(B) Fringe width

(C) Fresnel's distance

(D) Angular speed of central maximum

List-II

Formula

(I) $2\lambda/a$

(II) a^2/λ

(III) $2D\lambda/a$

(IV) $D\lambda/d$

Choose the correct answer from the options given below:

1. (A) - (III), (B) - (IV), (C) - (II), (D) - (I)
2. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
3. (A) - (II), (B) - (I), (C) - (IV), (D) - (III)
4. (A) - (IV), (B) - (III), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.15 Arrange the following in the increasing order of their energy gap between their conduction and valence bands:

- (A) Si
- (B) C (diamond)
- (C) Ge
- (D) Sn

Choose the correct answer from the options given below:

1. (D), (C), (A), (B)
2. (A), (C), (B), (D)
3. (B), (A), (D), (C)
4. (C), (B), (D), (A)

Options 1. 1

2. 2
3. 3
4. 4

Q.16 A photosensitive surface is illuminated by a point source of light 1 m away. When the source is shifted to 2 m, then

1. each emitted electron carries one eighth of the initial energy.
2. number of emitted electrons is half the initial number
3. each emitted electron carries half the initial energy
4. number of emitted electrons is a quarter of the initial number

Options 1. 1

2. 2
3. 3
4. 4

Q.17 The electric field vector of a plane electromagnetic wave oscillates sinusoidally at a frequency of 4.5×10^{10} Hz in a medium with permittivity ϵ_0 and permeability μ_0 . The wavelength of the wave is

1. 6.67×10^{-4} m
2. 2×10^{-3} m
3. 6.67×10^{-3} m
4. 6.67×10^{-12} m

Options 1. 1

2. 2
3. 3
4. 4

Q.18 It is observed that only 25% of a given radioactive sample is left undecayed after 16 days. What is the decay constant of the sample in day^{-1} ?

1. 0.25
2. 0.087
3. 8
4. $1/8$

Options 1. 1

2. 2
3. 3
4. 4

Q.19 Displacement current goes through the gap between the plates of a capacitor when the charge of the capacitor is

1. increasing only
2. decreasing only
3. increasing or decreasing
4. zero

Options 1. 1

2. 2
3. 3
4. 4

Q.20 Two long and parallel straight wires A and B carrying currents of 10.0 A and 5.0 A in the same direction are separated by a distance of 5.0 cm. Estimate the force on a 10 cm section of wire A.

1. 2×10^{-5} N
2. 2×10^{-4} N

$$3.10^{-5} \text{ N}$$

$$4.2 \times 10^{-3} \text{ N}$$

Options 1. 1

2. 2

3. 3

4. 4

Q.21 A coil of 400 turns of effective area 0.05 m^2 is kept perpendicular to a magnetic field of $5 \times 10^{-3} \text{ T}$.

When the plane of coil is rotated by 90° around any of its co-planar axis in 0.2 s in magnetic field, the emf induced in the coil is

1. 0.2 V

2. 0.05 V

3. 0.1 V

4. 0.5 V

Options 1. 1

2. 2

3. 3

4. 4

Q.22 The energy of a photon is equal to the kinetic energy of a proton. The energy of the photon is E . Let λ_1 be the de-Broglie wavelength of the proton and λ_2 be the wavelength of the photon. The ratio λ_1/λ_2 is proportional to the following

1. E^0

2. $E^{1/2}$

3. E^{-2}

4. E^{-1}

Options 1. 1

2. 2

3. 3

4. 4

Q.23 A galvanometer having a resistance of 7Ω is shunted by a resistance of 3Ω .

If the total current is 1 A , the value of current passing through the shunt is

1. 1.0 A

2. 0.7 A

3. 0.5 A

4. 0.3 A

Options 1. 1

2. 2

3. 3

4. 4

Q.24 Identify the correct statements among the following:

(A) Gauss's law is valid for the closed surface only.

(B) It is not possible to derive Gauss's law from Coulomb's law

(C) The electric field inside a uniformly charged thin spherical shell is zero

(D) The flux is contributed by the electric charges outside the closed surface

Choose the correct answer from the options given below:

1. (A), (B) and (D) only

2. (A), (C) and (D) only

3. (A) and (C) only

4. (B), (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.25 Kinetic energies of four particles a, b, c, d of equal masses are given below. Arrange them in order of their decreasing de-Broglie wavelengths.

(A) $(\text{KE})_a = 1 \text{ eV}$

(B) $(\text{KE})_b = 0.01 \text{ eV}$

(C) $(\text{KE})_c = 4 \text{ eV}$

(D) $(\text{KE})_d = 16 \text{ eV}$

Choose the correct answer from the options given below:

1. (A), (B), (C), (D)

2. (B), (A), (C), (D)

3. (B), (A), (D), (C)

4. (C), (B), (D), (A)

Options 1. 1

2. 2
3. 3
4. 4

Q.26 The two possible positions of an object kept in front of a lens of power +5.0 D, so that the image formed in both cases is four times magnified, are

- (A) -30 cm
- (B) -25 cm
- (C) -15 cm
- (D) -45 cm

Choose the correct answer from the options given below:

1. (A) and (D) only
2. (A) and (B) only
3. (C) and (D) only
4. (B) and (C) only

Options 1. 1

2. 2
3. 3
4. 4

Q.27 An unpolarized light of intensity I_0 is passed through a polariser and is incident on an analyser making an angle of $(\pi/6)$ rad with that of the polariser. The intensity of the light transmitted from the analyzer is

1. $0.33I_0$
2. $0.5I_0$
3. $0.375I_0$
4. $0.75I_0$

Options 1. 1

2. 2
3. 3
4. 4

Q.28 Three resistors, each of resistance R, are arranged in the following different combinations with a source of emf E.

Arrange them in the decreasing order of power dissipation:

- (A) Three resistors in series
- (B) Three resistors in parallel
- (C) One resistor in series with two resistors in parallel
- (D) One resistor in parallel with two resistors in series

Choose the correct answer from the options given below:

1. (D), (B), (A), (C)
2. (B), (D), (C), (A)
3. (A), (C), (D), (B)
4. (B), (C), (D), (A)

Options 1. 1

2. 2
3. 3
4. 4

Q.29 Consider the following statements:

- (A) An emf can be induced by moving a conductor in a magnetic field.
- (B) An emf can be induced by changing the magnetic field
- (C) An emf can be induced by changing the electric field
- (D) An emf can be induced by moving a conductor in an electric field

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A) and (B) only
3. (A) and (C) only
4. (B), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.30 On heating n-type semiconductors -

1. the number of holes increases while that of electrons decreases.
2. the number of electrons increases while that of holes decreases.
3. the number of holes and electrons remains same.
4. Both the number of holes and the number of electrons increase equally.

Options 1. 1

2. 2
3. 3
4. 4

Q.31 A battery of emf 12 V and internal resistance $3\ \Omega$ is connected to a resistor. If the current in the circuit is 0.8 A, the terminal voltage of the battery, will be

1. 12 V

2.15 V

3.9.6 V

4.2.4 V

Options 1. 1

2. 2

3. 3

4. 4

Q.32 A current of 10 A is flowing in a wire of length 1.2 m. A force of 12 N acts on it when it is placed in a uniform magnetic field of 2 T. The angle between the magnetic field and the direction of the current is

1. 30°

2. 45°

3. 60°

4. 90°

Options 1. 1

2. 2

3. 3

4. 4

Q.33 Which of the following statements is NOT correct?

1. The majority carriers in p - type semiconductors are electrons.

2. Doping pure germanium with pentavalent impurities gives n - type semiconductors.

3. The resistance of intrinsic semiconductor decreases with increase of temperature.

4. The conduction band have most electrons coming from the donor impurities.

Options 1. 1

2. 2

3. 3

4. 4

Q.34 At room temperature (25.0°C) the resistance of a heating element is $100\ \Omega$. The temperature of the element at which resistance becomes $125.5\ \Omega$ is : (Given that the temperature coefficient of the material of the resistor is $1.70 \times 10^{-4}\ ^\circ\text{C}^{-1}$)

1. 1525°C

2. 1500°C

3. 1475°C

4. $1500\ \text{K}$

Options 1. 1

2. 2

3. 3

4. 4

Q.35 A short bar magnet placed with its axis at 30° with a uniform external magnetic field of 0.16 T experiences a torque of magnitude 0.48 J. The magnetic moment of the bar magnet will be

1. 6 J/T

2. 0.4 J/T

3. 8 J/T

4. 1.5 J/T

Options 1. 1

2. 2

3. 3

4. 4

Q.36 In a parallel plate capacitor, each plate has an area of $7 \times 10^{-3}\ \text{m}^2$ and the distance between the plates is 2 mm. The space between the plates of the capacitor is filled with mica having dielectric constant 6. If the capacitor is connected to 120 V supply, the charge on each plate of the capacitor will be (Given: permittivity of the vacuum $\epsilon_0 = 8.85 \times 10^{-12}\ \text{SI unit}$)

1. $2.23 \times 10^{-10}\ \text{C}$

2. $2.23 \times 10^{-8}\ \text{C}$

3. $3.71 \times 10^{-9}\ \text{C}$

4. $3.71 \times 10^{-8}\ \text{C}$

Options 1. 1

2. 2

3. 3

4. 4

Q.37 The lateral shift produced by a glass slab during refraction through it, increases with -

(A) the increase in the thickness of the glass slab

(B) the increase in the value of the angle of incidence

(C) the increase in the value of the refractive index of the slab

(D) the increase in the length of the glass slab

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (A), (B), (C) and (D)
4. (B), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.38 An electric dipole of length 2 cm is placed at an angle of 30° with an electric field $3 \times 10^5 \text{ N C}^{-1}$. If the dipole experiences a torque of $9 \times 10^{-3} \text{ N m}$, the magnitude of either charge of the dipole is

1. $3\mu\text{C}$
2. 3 mC
3. $6\mu\text{C}$
4. 6 mC

Options 1. 1

2. 2
3. 3
4. 4

Q.39 A printed page is kept pressed by a glass cube ($\mu = 1.5$) of 5 cm edge. By what amount will the printed letters appear to be shifted when viewed from the top?

1. 2 cm
2. 1 cm
3. 1.7 cm
4. 2.5 cm

Options 1. 1

2. 2
3. 3
4. 4

Q.40 The separation between the plates of a charged parallel plate capacitor is increased. Which of the following quantities will change ?

- (A) charge on the capacitor plates
 - (B) potential difference across the capacitor
 - (C) energy of the capacitor
 - (D) energy density between the plates of the capacitor
- Choose the *correct* answer from the options given below:
1. (A), (B) and (D) only
 2. (A), (C) and (D) only
 3. (B) and (C) only
 4. (B), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.41 The half-life of a given radioactive nucleus is 200 days. The mean life of this nucleus is

1. 288 days
2. 2.88 days
3. 100 days
4. 200 days

Options 1. 1

2. 2
3. 3
4. 4

Q.42 For the L-R series circuit connected to an ac mains $e = e_0 \sin \omega t$, the average power consumed is:
(Symbols have their usual meanings)

1. $\frac{e_0^2 R}{2(R^2 + \omega^2 L^2)}$
2. $\frac{e_0^2 L}{2(R^2 + \omega^2 L^2)}$
3. $\frac{e_0^2}{2(R^2 + \omega^2 L^2)}$
4. $\frac{e_0^2}{2(R^2 + \omega^2 R^2)}$

Options 1. 1

2. 2
3. 3

Q.43 The phenomenon of splitting of white light into the constituent colours on passing through a prism is known as-

1. Refraction
2. Scattering
3. Dispersion
4. Diffraction

Options 1. 1

2. 2
3. 3
4. 4

Q.44 An infinite plane sheet of charge density 10^{-8} C m^{-2} is held in air. In this situation, the distance between two equipotential surfaces, whose potential difference is 10 V, will be

1. 17.7 mm
2. 4.42 mm
3. 11.7 mm
4. 8.85 mm

(Given: Permittivity of vacuum $\epsilon_0 \approx 8.85 \times 10^{-12} \text{ F/m}$)

Options 1. 1

2. 2
3. 3
4. 4

Q.45 The current in a coil falls from 5 A to 0 A in 0.1 s. If an average emf of 100 V is induced, then the self-inductance of the coil would be-

1. 1H
2. 4 H
3. 2 H
4. 5 H

Options 1. 1

2. 2
3. 3
4. 4

Q.46 Arrange the following series of emission lines in the spectrum of hydrogen atom in sequence according to their increasing wavelengths

- (A) Paschen series
- (B) Lyman series
- (C) Balmer series

Choose the correct answer from the options given below:

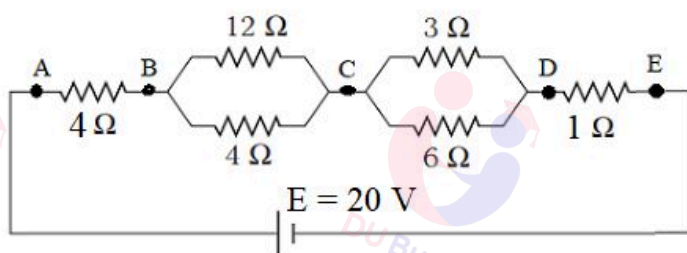
1. (A), (B), (C)
2. (B), (C), (A)
3. (C), (A), (B)
4. (C), (B), (A)

Options 1. 1

2. 2
3. 3
4. 4

Q.47

In the circuit diagram shown in the figure, the potential difference E is 20 V.



- (A) The potential difference across points C and D is V_{CD}
 (B) The potential difference across points B and C is V_{BC}
 (C) The potential difference across points D and E is V_{DE}
 (D) The potential difference across points A and B is V_{AB}

If the potentials V_{AB} , V_{BC} , V_{CD} , V_{DE} are arranged in increasing order, then Choose the **correct** answer from the options given below:

1. (C), (A), (B), (D)
2. (A), (B), (D), (C)
3. (B), (A), (D), (C)
4. (D), (C), (B), (A)

Options 1. 1

2. 2
3. 3
4. 4

Q.48 A charged particle carrying charge $q = 1\mu\text{C}$ moves in a uniform magnetic field with velocity $v_1 = 10^6\text{ m/s}$ at an angle of 45° with the x-axis in the xy-plane and experiences a force $\vec{F}_1 = 0.4\sqrt{2}\text{ N}$ along the negative z-axis. When the same particle moves with velocity $v_2 = 10^6\text{ m/s}$ along the z-axis, it experiences a force $\vec{F}_2$ in the y-direction. Find the magnitude of the magnetic field

1. 0.8 T
2. 8 T
3. 0.08 T
4. 8 mT

Options 1. 1

2. 2
3. 3
4. 4

Q.49 Which of the following will increase the mobility of charge carriers?

1. decrease in the charge of the mobile charge carriers
2. increase in the mass of the mobile charge carriers
3. increase in their average collision time
4. increase in the electric field

Options 1. 1

2. 2
3. 3
4. 4

Q.50 An unpolarised beam of light of intensity I_0 falls on a single polaroid. The intensity of the emergent beam is

1. $I_0/2$
2. I_0
3. $I_0/4$
4. Zero

Options 1. 1

2. 2
3. 3

